Market Regime Detection with PCA and K-Means:
Can an Unsupervised Cross-Asset Pipeline Recover Regime Structure That Carries Out-of-Sample Information About Future Risk?

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Abstract

Market environments alternate between latent "regimes" (calm trends, drawdowns, and high-volatility stress), but these states carry no labels. Practitioners organise them with the Bridgewater four-quadrant (growth × inflation) framework, yet prior unsupervised studies tend to collapse to roughly two regimes, a tension this note takes as its starting point. I ask whether a simple unsupervised pipeline (standardisation → PCA → K-means), fit on cross-asset prices, can recover a richer regime structure, and I adopt $k=4$ from the four-quadrant prior rather than the $k=2$ that cluster geometry alone favours. Eight liquid ETFs (equities, duration, gold, commodities, credit, and the VIX) are mapped to 43 leakage-free rolling features; the pipeline is fit only on 2010–2019, frozen, and applied to unseen 2020–2026 data. Out of sample the four regimes are ordered identically to training, and the stress regime carries $\sim 2.4\times$ the future 5-day volatility of the calm regime (27.1% vs. 11.4% annualised). The $k=4$ partition also separates future returns roughly seven times more than $k=2$, indicating the two extra regimes are economically real rather than noise. Whether the four recovered regimes coincide with the four macro regimes of the prior is examined and left partly open, motivating a synthetic-data recovery test as the natural next experiment.

1 Introduction

Asset returns are close to unpredictable, but the *environment* in which they are generated is not static: markets move through persistent states in which volatility, correlations, and tail risk behave very differently. Identifying these states matters in practice, because a portfolio's risk depends heavily on the prevailing regime, and it is interesting scientifically, because the states are latent: no data feed publishes a column labelled "today = stress." The central question of this note is whether a purely unsupervised pipeline can *recover* such regime structure from cross-asset prices, and whether the recovered structure is rich enough to match how practitioners actually think about regimes. The headline result is that a frozen pipeline separates future risk roughly $2.4\times$ between its calmest and most stressed states on data generated after everything it learned from.

1.1 Related Work

Regime detection is a mature field, and two patterns recur across the unsupervised literature. First, the inputs are usually narrow: Hamilton's Markov-switching model [2] operates on a single macro or return series, and change-point and clustering methods are often applied to one asset's volatility series at a time [7]. Second, coarse or purely geometric methods tend to recover only about two states, a clean "calm versus stress" cut. The work closest to ours is Akioyamen et al. [1], who apply a PCA $\to$ K-means pipeline, but to US macroeconomic statistics rather than a cross-asset price panel. Statistical jump models have been used on equity-index returns for downside-risk control [3], and regime persistence is well documented in the switching-model literature [6]. The methodological ingredients are therefore established; what varies is the input and the number of regimes the method is allowed to express.

1.2 Motivation and Hypothesis

The gap this note steps into is a tension between data geometry and macro theory. Purely geometric model selection on regime features tends to prefer $k=2$ (calm versus stress), but that two-state answer is inconsistent with the framework practitioners rely on. The Bridgewater four-quadrant framework [6, 8] organises markets by growth × inflation into *four* regimes (loosely: rising growth with falling inflation; rising growth with rising inflation; falling growth with rising inflation, that is stagflation; and falling growth with falling inflation). A method that can only express two states cannot represent this structure. I therefore treat the number of regimes as a hypothesis rather than a quantity to read off a silhouette curve, and I adopt $k = 4$ *because* that is the count the four-quadrant prior posits, then test whether the data reward it.

Hypothesis. *A PCA $\to$ K-means pipeline, fit only on* 2010–2019 *and then frozen, recovers cross-asset regimes whose differences in future risk persist out-of-sample on* 2020–2026 *data.* Three sub-questions follow, each tied to the tension above: (i) does the data support $k=4$ over the geometric default $k=2$; (ii) are the four regimes economically interpretable, and do they correspond to the four macro regimes of the prior, each with a distinct economic background; and (iii) does the structure generalise out-of-sample? A broad cross-asset price panel (equities, bonds, gold, commodities, credit, and the VIX) is what makes (ii) even askable: it lets the model represent flight-to-safety, that is, capital rotating out of equities into bonds and gold, which a single-index or pure-macro study structurally cannot.

2 Methodology

Features. Let $P_t^{(a)}$ be the adjusted close of asset a and $r_t^{(a)} = \ln\bigl(P_t^{(a)}/P_{t-1}^{(a)}\bigr)$ its daily log return. For each tradeable asset I form trailing-window features

$$R_t^w = \sum_{i=0}^{w-1} r_{t-i}, \qquad \sigma_t^w = \operatorname{std}\bigl(r_{t-w+1:t}\bigr)\sqrt{252}, \tag{1}$$

$$D_t^w = \frac{P_t}{\max_{t-w<j\le t} P_j} - 1, \qquad \rho_t = \operatorname{corr}_{60}\bigl(r^{\mathrm{SPY}},\, r^a\bigr), \tag{2}$$

with return windows $w \in \{5, 20, 60\}$ and volatility windows $w \in \{20, 60\}$. The full set is 7 assets $\times(3+2) = 35$, plus two SPY drawdowns, two VIX features (level and 5-day change), and four SPY cross-asset correlations, for 43 features. All windows are trailing; the only forward-looking quantities are evaluation targets (future 5/20-day return and volatility), which live in a separate frame the pipeline never receives.

Pipeline. Features are standardised, since the scales are incomparable (VIX ≈ 20, a correlation $\in [-1, 1]$), and without scaling the VIX column would dominate every distance. The standardised matrix $X \in \mathbb{R}^{n\times 43}$ is then reduced by PCA, which diagonalises the sample covariance $\Sigma = \frac{1}{n}X^\top X = V\Lambda V^\top$ and projects each day onto the leading eigenvectors $V_{1:d}$, retaining the smallest d for which the explained variance $\sum_{i\le d}\lambda_i/\sum_i \lambda_i \ge 0.90$. The pipeline is:

$$\underbrace{\text{StandardScaler}}_{\text{equal weight}} \;\rightarrow\; \underbrace{\text{PCA}(0.90)}_{d=16\text{ components}} \;\rightarrow\; \underbrace{K\text{-means}(k{=}4)}_{n_{\text{init}}=50}.$$

Retaining 90% of training variance yields $d = 16$ orthogonal components. This matters because the raw features are highly redundant: the 12 volatility columns are ~ 0.9 correlated, so clustering on raw inputs effectively feeds one "volatility" signal a dozen times and drowns out credit, gold, and momentum structure. Each day becomes a point $z_t \in \mathbb{R}^{16}$, and K-means picks centroids $\{\mu_k\}$ minimising within-cluster inertia

$$J = \sum_t \bigl\| z_t - \mu_{c(t)} \bigr\|_2^2, \tag{3}$$

with $n_{\text{init}} = 50$ restarts (the objective is seed-sensitive, so the lowest-inertia fit is kept) and a fixed random seed. The pipeline outputs only an integer label 0 to 3 per day; names are assigned afterward.

Interpretability. Because the components are linear combinations of named features, their loadings admit

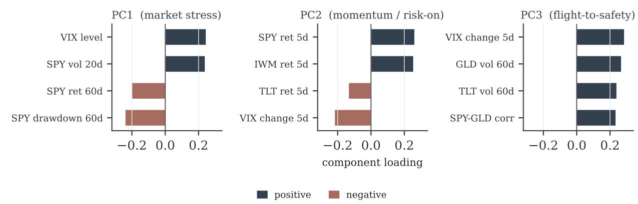

Figure 1: Largest standardised loadings of the first three principal components. The leading factors map onto recognisable economic axes; later components are not assigned labels.

economic readings (Fig. 1). PC1 (market stress) loads positively on the VIX level and SPY 20d volatility and negatively on SPY drawdown and trailing return; PC2 (momentum / risk-on) loads positively on short-horizon equity returns and negatively on the VIX change and Treasury returns; PC3 (flight-to-safety) loads positively on the VIX change, gold and Treasury volatility, and the SPY-gold correlation. Later components capture finer variation, resist clean labels, and are left unnamed.

3 Data and Experimental Design

Universe. Eight instruments span the axes that define a regime: equity beta (SPY, QQQ, IWM), duration (TLT), real assets (GLD, DBC), credit (HYG), and forward-looking fear (VIX). Daily adjusted closes are pulled from Yahoo Finance via the `yfinance` package [9] with `auto_adjust=True` (splits and dividends baked in) from Jan 2010 (post-GFC; the date all eight already trade) through May 2026, giving a clean aligned panel. The data are free and keyless, so the pipeline is fully reproducible.

Chronological split. The split is strictly temporal and never shuffled, since shuffling would scatter future days into training and leak information. The training set spans 2010 to 2019 (2,456 trading days, including the taper tantrum and the 2018 Q4 selloff); the test set spans 2020 to 2026 (1,587 days, including the COVID crash, the 2022 inflation selloff, and the 2023 banking scare). PCA and K-means are fit on training rows only and applied unchanged to test rows, making the test a genuine forecast: the model is judged on data generated *after* everything it learned from.

Controlled comparisons. Two comparisons isolate each design choice (Sec. 4): $k = 4$ vs. $k = 2$, and PCA vs. raw features. A non-ML VIX-quartile baseline tests whether the multi-asset model adds anything beyond a single volatility number.

4 Results

Choosing k. The silhouette coefficient of point i is $s_i = (b_i - a_i)/\max(a_i, b_i)$, where a_i is its mean distance to its own cluster and b_i the mean distance to the nearest other cluster; the mean over points scores a partition

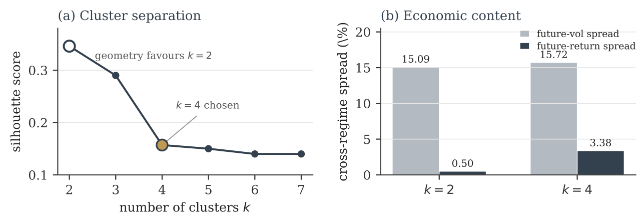

Figure 2: Model selection for k. (a) The silhouette score decreases monotonically, so a purely geometric criterion favours $k=2$. (b) On an independent economic criterion, $k=4$ separates future returns roughly seven times better while matching $k=2$ on future volatility, indicating the two extra clusters are real structure rather than noise.

Table 1: Training-period regime profiles. Names assigned from these statistics; K-means emits only cluster IDs.

Regime	VIX	20d ret	20d vol	Days
Calm	13.4	+1.6%	9.3%	1153
Choppy / transition	16.4	+2.8%	12.9%	761
Risk-off / drawdown	22.2	−3.7%	18.5%	396
Stress (high-vol)	30.3	−1.0%	30.1%	146

geometrically. This score is monotone-decreasing in k and favours $k = 2$ (0.35 vs. 0.16 at $k = 4$), and inertia falls smoothly with no sharp elbow (Fig. 2a). Geometry alone therefore prefers exactly the two-state cut that the macro prior rejects. Holding to the $k=4$ hypothesis from Section 1.2, I tested whether the data reward it. They do: while $k=2$ and $k=4$ separate future *volatility* about equally (15.1% vs. 15.7%), $k=4$ separates future *returns* $\sim 7\times$ better (3.38% vs. 0.50%; Fig. 2b). The two extra clusters split a low-return "risk-off drawdown" state from a high-volatility "stress" state, two states with opposite forward returns that $k=2$ merges.

Discovered regimes. Names are assigned post-hoc from training statistics (Table 1). Critically, the two "bad" states are distinct: *Risk-off* has the worst returns (−3.7%) while *Stress* has the highest volatility (30%) and deepest drawdown (−8.7%).

Main out-of-sample result. On held-out 2020–2026 data the regime ordering by future risk is preserved exactly: future 5-day volatility rises monotonically from Calm (11.4%) to Stress (27.1%), a $2.4\times$ gap (Fig. 3). Because the pipeline was frozen in 2019, this is direct evidence the learned structure generalises rather than fitting in-sample idiosyncrasies; on the SPY price path the frozen model flags the 2020 COVID crash and the 2022 inflation selloff as Stress/Risk-off days it was never shown.

Robustness. The comparisons (Table 2) decompose the design. Dropping PCA leaves the spreads essentially unchanged but lowers the silhouette and forfeits

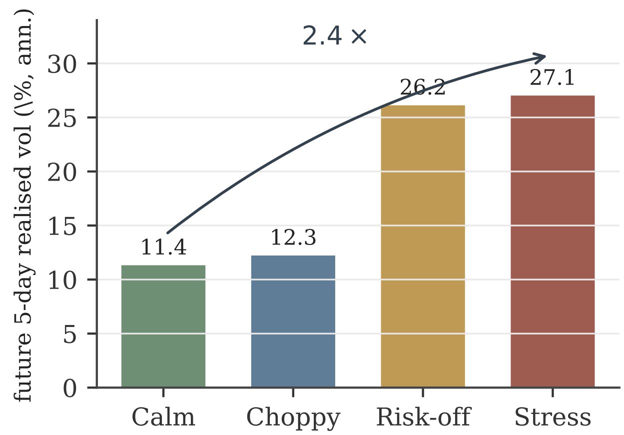

Figure 3: Out-of-sample future 5-day realised volatility by regime (2020–2026). Ordering is preserved from training; Stress carries $\sim 2.4\times$ the future volatility of Calm.

Table 2: Comparisons over k and PCA (train silhouette; spreads on future risk).

Variation	Silh.	Vol sp.	Ret sp.
PCA + KM (k=4)	0.157	15.7%	3.38%
PCA + KM (k=2)	0.346	15.1%	0.50%
Raw + KM (k=4, no PCA)	0.137	15.8%	3.44%

the interpretable factors, so PCA stays for interpretability rather than raw separation. Against the non-ML baseline, VIX quartiles separate pure future volatility slightly better (17.1% vs. 15.7%), unsurprising since the VIX is itself a volatility forecast; the clustering's added value is the return dimension and the risk-off-versus-stress distinction a single number cannot express.

Behavioural check. As an illustration (*not* a deployable strategy), holding 50% SPY on predicted-Stress days and 100% otherwise over the test period lowers annualised volatility from 20.5% to 17.9% and maximum drawdown from −35.8% to −30.1%, at the expected cost of return (12.6% → 9.6%): the Stress label de-risks exactly when realised risk is elevated.

5 Discussion

Interpreting the result. The findings line up with theory rather than contradicting it. Returns separate far less cleanly than risk, the empirical signature of the efficient-markets view combined with volatility clustering [4, 5]: direction is close to a martingale, but conditional variance is persistent and therefore learnable. This is why the note targets future risk and does not advance a return-prediction claim, a scope decision rather than a weakness. The preserved out-of-sample ordering indicates the clusters capture a stable property of the volatility process rather than a one-decade artefact, consistent with regime persistence in the switching-model literature [2, 6]. The disciplined model-selection conflict is the methodological core: a purely geometric criterion (silhouette) prefers $k = 2$, yet that cut cannot tell a sharp crash from a slow drawdown, so I treated silhouette as necessary but not sufficient and adjudicated k on an independent economic criterion. A high internal-validity metric should not override out-of-sample economic content.

Do the four regimes match the macro prior? This is the most important open question, and it should not be overstated. I adopted $k = 4$ because the Bridgewater framework posits four growth $\times$ inflation regimes, but adopting that *count* does not establish that the four clusters the model finds *are* those four economic regimes. The recovered regimes are ordered by risk (Calm, Choppy, Risk-off, Stress), which is a volatility and drawdown taxonomy, not directly the growth-versus-inflation taxonomy of the prior; there is no Goldilocks or stagflation label in the data, only cluster IDs that I named after the fact. The cross-asset panel makes a correspondence *plausible* (commodities and gold proxy inflation; duration and equities proxy growth), but plausibility is not verification. Asserting four regimes and finding four interpretable clusters is therefore consistent with, but not proof of, the four-quadrant structure; the clusters could equally be a risk-ordered partition that merely happens to have four levels. Settling this requires labels the data do not provide.

The ground-truth problem and a recovery test. On real data there is no ground truth: nothing tells us whether four is the right number, or whether the same four regimes would reappear on another dataset. The clean way to test recovery is to remove the ambiguity by construction. One could generate synthetic but realistic market data driven by a known, deterministic regime process (for example a hidden Markov process over growth and inflation states with regime-dependent return and volatility dynamics), apply the frozen pipeline, and measure how well the recovered clusters align with the planted labels. That experiment would convert the present interpretive claim into a falsifiable one and is the natural next step.

Limitations. Several caveats bound the claim. K-means assumes spherical, equal-size clusters; a Gaussian Mixture or Hidden Markov Model would represent regime shape and persistence more naturally and would also supply transition probabilities. Overlapping rolling windows make consecutive days autocorrelated, so naive significance tests overstate confidence and a block bootstrap or Newey-West correction is needed for formal p-values. Regime names are human and post-hoc. The test window is dominated by a few extreme events (COVID, 2022), limiting out-of-sample regime diversity. The behavioural check ignores transaction costs, slippage, and execution lag and is illustrative only.

Conclusion. Trained on one decade and frozen, a PCA $\rightarrow$ K-means pipeline recovered four cross-asset regimes whose ordering by future risk persisted out-of-sample, with the stress regime carrying roughly 2.4$\times$ the future volatility of the calm regime. The number of regimes was set by a macro prior rather than by geometry, and the data rewarded that choice on an independent return criterion. Whether the four clusters are the four economic regimes of the prior, rather than a risk-ordered partition of comparable granularity, remains the open question a synthetic-data recovery test is designed to answer.

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